Geometrically Consistent Generalizable Splatting

Abstract

Gaussian splatting has emerged as the preferred 3D scene representation due to their incredible speed and accuracy in novel view generation. Various attempts thus have been made in adapting multi-view structure prediction networks to directly predict 3D-splats from images. However, most of the existing work in this domain have focused on enhancing existing to view structure predictors to estimate additional Gaussian Splats perameters – orientations, scale, opacity and appearance, by minimizing view-synthesis loss. We show that such view-synthesis loss is insufficient to learn geometrically meaningful splat representations. We systematically analyse and address the inherent ambiguities in 3D splats in representing structure and appearance of the scene in the learning process. We present a novel framework for to learn networks for self-supervised, pose-free generalisable splatting, that explicitly ensures geometric correctness . By, incorporating geometric consistency into learning, our proposed approach provides generalizable splats that provides (i) state of the art 3D reconstructions, (ii) two-view pose estimation and (iii) in domain novel-view synthesis on RealState10K dataset. We also showcase zero-shot capabilities of the proposed generalizable splat network indoor dataset ScanNet, where the proposed method outperforms the prior art substantially on geometric tasks.

1 Introduction

Gaussian splatting [**?**] has revolutionized 3D structure and appearance modelling from multiple 2D images in the recent past. Departing from traditional depth or point cloud representations for structure of the scene, 3D Gaussians allows to model both scene structure and appearance to provide high-fidelity 3D reconstruction from multiple views. 3D Gaussians implicitly model surface reflections and refractions properties and environment lighting to encode view dependent scene appearance, is memory efficient representation when compared to explicit volumetric alternatives. They facilitate rendering of the scene form arbitrary viewpoints in fraction of a second utilizing modern GPUs.

Owing to this success of 3DGS, a significant effort has been put to improve the 3DGS approach to work with few-views [], in presence of dynamically moving objects [] and in incorporating object semantics into 3D reconstructions []. Real-time simultaneous localization and mapping approaches have been adapting splats as the inherent scene representation []. Splats are also used for generation of geometrically consistent images and video sequences []. 3D Gaussian splats have become the prevalent choice for representing scenes in this era.

The deep-learning revolution of the last decade has drastically influenced geometric inference from one or multiple images as well. While earlier works in this direction focused on learning neural networks that map image(s) to their depth-maps provided by range sensors [**?**], recent works has explored reconstructing registered set of per-pixel point clouds for multiple images [**?**] to provide state-of-the-art relative pose and structure of the scene. It has also been shown that these feed-forward neural networks to predict geometry from image(s) can be trained without depth sensors, in a self-supervised manner, by minimizing view synthesis loss [**? ? ? ? ?**]. Structure prediction from single or few images have been used as the optimization-free building block in high-fidelity tracking and mapping systems []. Learning assisted recovery of scene geometry has truly become mainstream.

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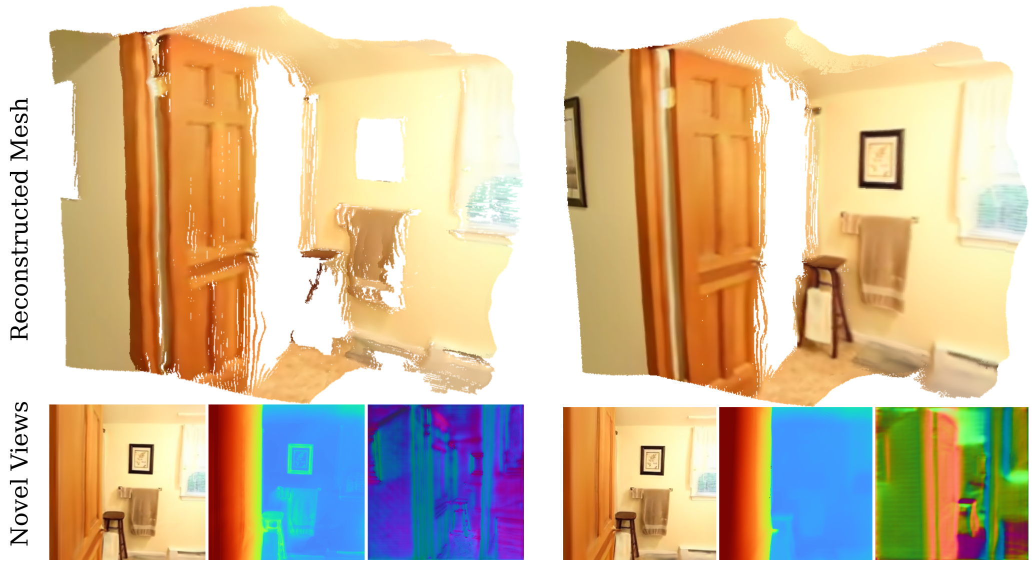


(a) NoPoSplat (b) Ours

Figure 1: We reconstruct the mesh of the scene (1^{st} *row*) from the splats using truncated signed distance function (TSDF) to fuse the rendered novel-view RGB images and depthmaps (2^{nd} *row*). While NoPoSplat produces reasonably good appearance quality in the rendered RGB, it fails to learn accurate geometry, resulting in multiple holes in the reconstructed mesh due to the inconsistencies across the depthmaps during TSDF fusion.

It is natural than, to marry the incredible capabilities of modern deep neural networks in learning useful geometric priors with the advantages of using 3D Gaussians as the scene representation. Various laudable attempts have been made recently, in training neural networks to predict 3D splats directly from images [**? ? ? ? ? ?**]. These methods are usually dubbed as Generalizable Gaussian Splatting methods in the literature. Most Genralizable Gaussian Splatting methods, focus on designing or adapting structure prediction networks to estimate means of the Gaussian. Generally these networks consist of encoders for one or multiple images and decoders to predict geometry – usually in the form of depth or point cloud. Additional decoder layers are usually appended to decoders additionally predict the remaining Gaussian parameters (orientation, scale, opacities and view-dependent color). All or a partial set of neural network parameters are than learned by usually minimizing simple view-synthesis loss on a few target views by several generalizable splatting methods.

This prevalent setup used by Generalizable Gaussian Splatting methods overlooks several key issues that are inherited from underline 3DGS optimization that is used for supervision by almost all aforementioned methods:

- Splats are grossly over-parametrized compared to traditional depth-maps or 3D point-clouds used for few view reconstruction. Successful estimation of 3D splats usually require a large number of densely sampled viewpoints. 3DGS with few-views is an active field of research, with several solutions relying on heavy regularization []. These developments are ignored in building a successful generalizable Gaussian splatting technique.
- Unlike per-pixel depths or 3D locations, that are uniquely defined for a scene (upto scale), numerous configurations of the 3D Gaussians can explain scene structure and appearance equally well. In absence of a unique definition for the optimal set of Gaussians, it is unclear how to learn 3D Gaussians even when scene structure is available for supervision in the form of depth-maps.
- Successful Gaussian splatting rely on multiple non-differentiable heuristics (such as pruning, splitting and duplication of Gaussian). Over the course of the optimization, many Gaussians die (reach close to zero opacity), or are introduced late in the optimization. In absence of differentiable approximations of these heuristics existing generalizable splatting approaches

hope that the inductive biases in network design and large datasets will help keeping the Generalized Gaussians perpetually active during the training for stochastic gradient decent to work.

As result of these oversights, the current generalizable Gaussian splatting solutions often converge to geometrically degenerate scenarios. This is spacially true for Gaussian parameters other than their means. The means of the Gaussians are relatively stable thanks to the well studied problem of self supervised single of two-view depth estimation methods.

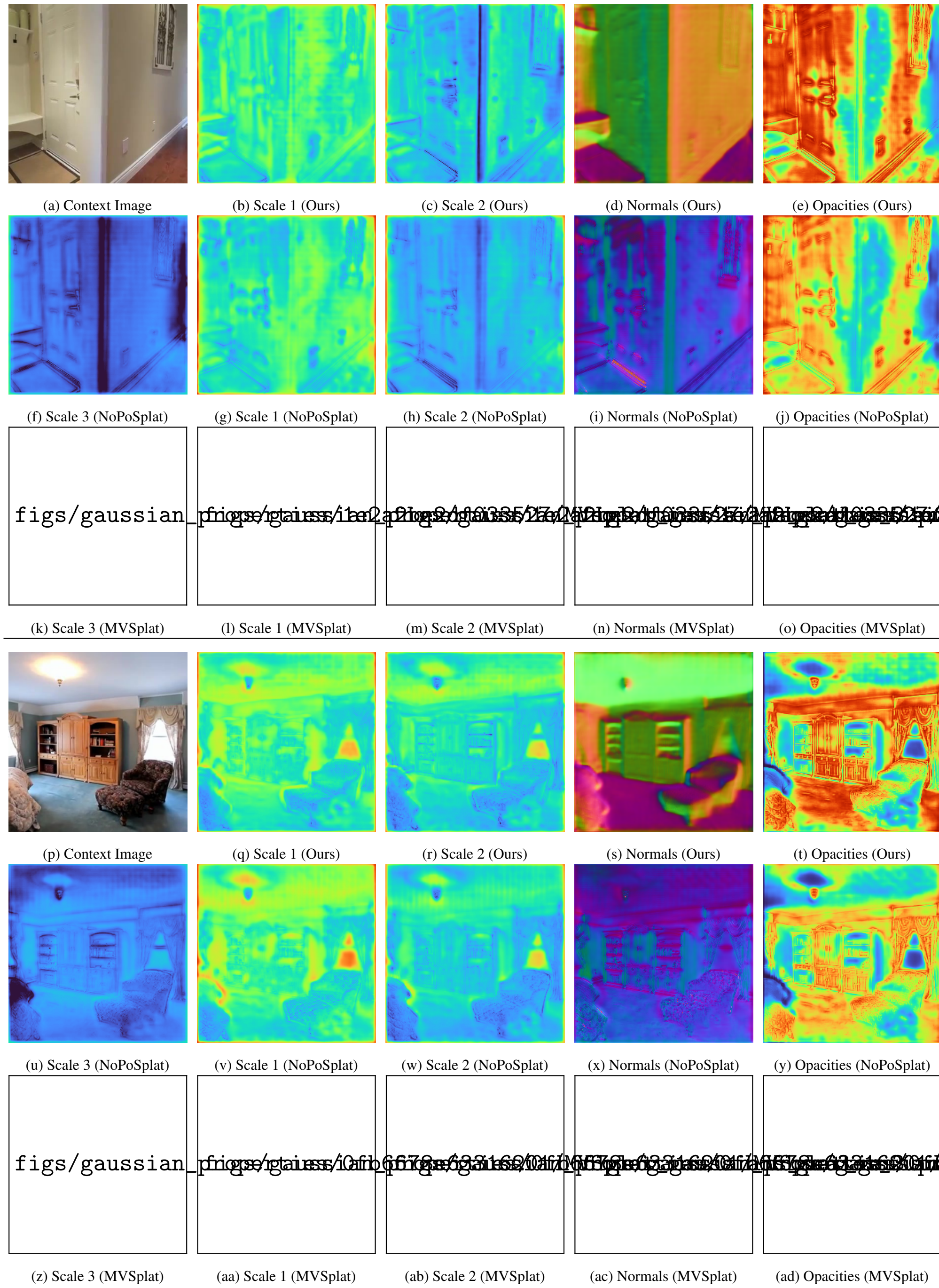
In Figure 2, we showcase how current generalizable splatting frameworks fail to learn meaningful opacities, orientations or scale of the Gaussians. These frameworks often end up over-fitting to single task of novel-view synthesis – which is used for learning Gaussian parameters. The plausibility of 3D Gaussians configurations learned by these methods is questionable and no geometric meaning can generally be inferred from the these predicted Gaussians directly. On the contrary, many interesting geometric properties are reflected directly in the visualization of the Gaussian opacity, scales and orientation the proposed method predicts. Briefly, one can see that proposed method reconstruct very low number of splats with small opacities (near dead Gaussians) compared to baselines, can provide correct surface normals directly from the predicted Gaussian orientations, and produce Gaussian shapes elongated perpendicular to the normal edges and very small scale along edges. We will extensively discuss the geometric inconsistency present in the current Generalizable 3DGS results and how proposed method alleviate these inconstancies in section **??**.

The two approaches we picked for visualization of the Gaussian quantities broadly represent separate underline representation for encoding Gaussian means - namely (i) per-pixel depthmaps or (ii) per-pixel 3D locations aligned in a single frame of reference. However, to our knowledge, all the Generalizable 3DGS methods suffer from the same limitations.

In this work, we systematically aim to define what the ideal configuration for the geometrically consistent Gaussian should looks like and proposed appropriate priors to assist Generalized Gaussian Splatting. To that end we opt to work on the recently published NoPo-Splat [**?**] as our baseline and enhance it. We select No-Posplat as baseline because, aside from its state of performance on multiple geometric tasks, it is a self-supervised approach that does not require ground truth depth-maps unlike some of the other generalizable splatting adaptions[**?**]. Additionally, utilizing DUST3R framework, the approach is one of the few to provide generalizable splatting from a pair of images without requiring relative pose of these images. However, the addition ambiguities in using splats as representation poses similar problems in learning geometrically accurate Genralizable Gaussian Splatting and contribution of our work can help other alternative baselines as well.

We systematically address these ambiguities by carefully changing the scene representation and adding suitable regularizations to the tradinally used view synthesis loss. Our main observations are, (i) Defining the ideal Gaussian orientations to be dominant normals of the surfaces they are part of (as advocated in [] help resolving structural ambiguities. We do that be intergrading 2D Gaussian splatting restarized used in [] in place of the commonly used 3D Gaussian Resterizer [] (ii) when relying on view synthesis loss, ensuring that Gaussians are pixel-aligned is extremely important. In particular, for the pose free methods that aim to use PnP style initialization for relative pose estimation, it is a crucial prior to introduce (iii) jointly learning pixel-aligned normals and 3D location is challenging when self supervision in used. It is best to rely on depth-derived pseudo-normals to supervise for Gaussian orientations for accurate normal predictions.

We train presented generalizable splatting networks on RealState10k dataset [] and show that we outperform the NoPoSplat baseline on the tasks of Novel-View Synthesis, Geometric Correctness and relative pose estimation on RealState10k testset. We also showcase generalization capability of our trained models on Scan-net [] and DTU datasets []. We outperform all existing methods in terms of relative-pose estimation form two-images by a large margin on all the dataset we test our approach on. This includes methods that are tailored specifically to estimate relative pose such as [**? ?**]. On all datasets, our approach provides state of the art view synthesis when compared to pose-free generalizable splatting and pose-free nerf alternatives. Our approach provides most accurate 3D reconstructions amongst all generalizable splatting approaches in the zero-shot testing setup on DTU and Scan-net datasets including methods that use ground truth relative pose. To the best of our knowledge, it is the first generalizable splatting approach to analyse and evaluate the veracity and meaningfulness of Gaussian Splatting parameters such as opacities and orientation. We believe that the presented analysis pave the way forward for future research in the domain.



(a) Context Image (b) Scale 1 (Ours) (c) Scale 2 (Ours) (d) Normals (Ours) (e) Opacities (Ours)
(f) Scale 3 (NoPoSplat) (g) Scale 1 (NoPoSplat) (h) Scale 2 (NoPoSplat) (i) Normals (NoPoSplat) (j) Opacities (NoPoSplat)
(k) Scale 3 (MVSplat) (l) Scale 1 (MVSplat) (m) Scale 2 (MVSplat) (n) Normals (MVSplat) (o) Opacities (MVSplat)
(p) Context Image (q) Scale 1 (Ours) (r) Scale 2 (Ours) (s) Normals (Ours) (t) Opacities (Ours)
(u) Scale 3 (NoPoSplat) (v) Scale 1 (NoPoSplat) (w) Scale 2 (NoPoSplat) (x) Normals (NoPoSplat) (y) Opacities (NoPoSplat)
(z) Scale 3 (MVSplat) (aa) Scale 1 (MVSplat) (ab) Scale 2 (MVSplat) (ac) Normals (MVSplat) (ad) Opacities (MVSplat)

Figure 2: Qualitative comparison of Gaussians. First row visualizes our 2D Gaussians with the second and third containing results form NoPoSplat and MVsplat respectively. All visualized Gaussians correspond to the context image (a). Row 1: (b-c) shows extent of 2D Gaussians along the estimated scene surface it belongs to. The surface is defined by its normal which is shown in (d). Row 2/3 for NoPosplat and Mvsplat respectively: (f/k) show minimal Gaussians elongation. This direction is assumed to be perpendicular to the dominant surface and the elongation quantizes the uncertainty in the surface localization. (g/l, h/m) show Gaussian elongation along the estimated dominant surface, i.e. comparable to corresponding columns in top row, (i/n) in the two rows shows estimated normals of the dominant surface for baseline methods. Additionally, (e,j,o) show the estimated opacity of the Gaussians.

2 Related Work

todo

2.1 Self-supervised two view / single view reconstruction

todo

2.2 3D and 2D GS

todo

2.3 Genralizable splats : an overview

From Intro (maybe fits here better) The first set of methods [1**?** **?** , 2] (represented by MV-splat[2] results in figure 2). These methods uses image depths as an internal representation for Gaussian centres. These centres are computed by directly back-projecting pixels of image with the help of known intrinsics and estimated depths. These Gaussians are thus explicitly aligned with pixels of the input (or context) image . Note that many of these methods rely on the reparametrization trick proposed in the pixel-splat [1] framework where opacities are considered proportional to the discrete depth probabilities.

The second set of methods [**?** **?** **?**] are represented by NoPoSplat [**?**]. These methods use DUST3R [6] setup to predict Gaussian centres directly from images. More precisely, they predict the 3D locations of pixel-aligned Gaussians for every input image in a single canonical frame of reference – usually the first input frame. These methods usually estimate Gaussuin opacities as an additional output of a decoder. Canonical Gaussian centres are expected to be projected onto image pixels providing a way to estimate intrinsics and extrinsic camera parametersusing suitable objective function. These methods are also dubbed as "pose free" Genralizable 3DGS methods.

todo...

2.3.1 Methods that use explicit depth parametrization but need relative pose

todo

2.3.2 Methods that use point cloud and extend duster requiring no pose

3 Method

In this section, we introduce our geometrically consistent Gaussian splatting framework. We build upon the NoPoSplat architecture but propose a novel loss function and architectural refinements to enhance its geometric consistency and generalization capabilities. We first define the problem formulation and notation, briefly review the NoPoSplat architecture and learning framework, and then detail the novel losses and architectural modifications introduced in this work.

We assume we are given a sparse set of multi-view images, each with known camera intrinsics. We denote these images by $\mathcal{I} = \{\boldsymbol{I}^i\}_{i=1}^K$, where $\boldsymbol{I}^i \in \mathbb{R}^{H\times W\times 3}$, and their corresponding camera intrinsics by $\mathbf{K}^i \in \mathbb{R}^{3\times 3}$. The primary objective is to learn a feed-forward neural mapping $f_{\boldsymbol{\theta}}$, parameterized by $\boldsymbol{\theta}$, that predicts a set of 3D Gaussians representing both geometric structure and appearance attributes of the underlying scene:

$$f_{\boldsymbol{\theta}} : \{(\boldsymbol{I}^i, \boldsymbol{K}^i)\}_{i=1}^K \mapsto \left\{\left(\boldsymbol{\mu}_j^i, \alpha_j^i, \boldsymbol{q}_j^i, \boldsymbol{s}_j^i, \boldsymbol{c}_j^i\right)\right\}_{j=1,\dots,H\times W}^{i=1,\dots,K}, \tag{1}$$

where each Gaussian is characterized by its center position $\boldsymbol{\mu} \in \mathbb{R}^3$, orientation represented by a unit quaternion $\boldsymbol{q}$, and scale parameters $\boldsymbol{s}$, which can be either $\boldsymbol{s} \in \mathbb{R}^2$ for 2D Gaussians or $\boldsymbol{s} \in \mathbb{R}^3$ for 3D Gaussians. The appearance parameters consist of an opacity value $\alpha \in \mathbb{R}$ and spherical harmonics (SH) coefficients $\boldsymbol{c} \in \mathbb{R}^k$, with k degrees of freedom.

3.1 Pipeline

Our approach builds upon the architecture proposed by [**?**] and involves three main modules: a transformer-based encoder, a corresponding decoder, and specialized heads to predict Gaussian parameters. Notably, both encoder and decoder employ Vision Transformer (ViT) backbones exclusively, without introducing geometric constraints such as epipolar geometry or multi-view cost volumes. Despite this simplicity, the purely transformer-based design achieves performance comparable or superior to approaches relying on geometric priors, especially in scenarios where input views exhibit minimal overlap.

Specifically, the encoder transforms RGB images into patch-based sequences, which are flattened and augmented by camera intrinsic tokens. These token sequences are individually processed by a ViT encoder with shared parameters across views. The resulting encoded features are subsequently fused via cross-attention layers in the ViT decoder, effectively aggregating information across multiple views. Gaussian parameters are estimated by two separate prediction heads: one head directly infers Gaussian centres positions from decoder features alone, while the other head predicts appearance-related and geometric parameters using both decoder outputs and the original RGB images.

3.2 Geometric Consistency

In this part, we introduce novel losses and architectural modifications aimed at ensuring geometric consistency in the predicted feed-forward Gaussians. Transitioning from a full 3D Gaussian representation to a 2D surfel-based representation, as demonstrated by [**?**], seems a natural step toward improved geometric consistency. However, an important open question is whether this change in representation alone suffices to achieve consistent scene geometry in a feed-forward setup, while simultaneously preserving or even enhancing novel-view synthesis quality.

As shown in Section 5, although switching to surfel-based representations improves geometric consistency compared to the original 3D Gaussian baseline, it degrades the quality of novel-view synthesis, consistent with findings reported in [**?**]. To further enhance geometric consistency without sacrificing image quality, we propose two novel regularization terms specifically designed for the context of feed-forward generalizable splatting: one to guide the orientation of Gaussian surfels, and another to enforce their alignment with the pixel grid.

Alongside the standard RGB reconstruction loss $\mathcal{L}_{color}$—which integrates both an $\mathcal{L}_2$ and a perceptual LPIPS term—we introduce a normal consistency regularization $\mathcal{L}_{normal}$ to align Gaussian surfel orientations with rendered depth, and a grid alignment loss $\mathcal{L}_{grid}$ to ensure precise pixel-level alignment. These losses are combined into the final objective:

$$\mathcal{L} = \mathcal{L}_{color} + \lambda_n \mathcal{L}_{normal} + \lambda_g \mathcal{L}_{grid}, \tag{2}$$

where λ_n and λ_g are weighting factors balancing the influence of each regularization. Next, we provide detailed formulations and motivations for these novel losses.

3.2.1 2D-GS

Why 2DGS is better than 3DGS and how we change anything we had to. For example Scale prediction and rasterization change details goes here.

3.2.2 Learning Gaussian orientations

Normal loss goes here

3.3 Making 3D Gaussian pixel aligned

Grid loss and details go here

3.4 Fixing dead Gaussians

Activation / alpha loss ? Even if they dont improve we can write about options but we end up saying that previous changes fixes alpha

4 Analysis of figure 1

It is evident in Figure 2 that none of the existing Genralizable 3DGS approach provide meaningful orientations or scales for the estimated Gaussians. In particular, we visualize the normal of the dominant plane of 3D Gaussions that corresponds to the direction in which the Gaussians is minimally elongated. This minimal elongation is labelled as scale 3 in the figure and quantify the uncertainty in localising scene surface around this Gaussian. The estimated normals form the existing Generalizable 3DGS methods lack any geomtric meaning whereas the proposed method can provide meaningful normal-maps confirming to plaussible scene geometry. For example notice multiple planes confirming to Manhattan prior in the scene in Figure **??** and the corresponding normal predictions.

The existing Generalized 3DGS methods does not provide geometrically meaningful estimates of the shape (scales) to the Gaussians either. One can observe clear over-fitting to view synthesis loss – which is highly sensitive to location of image edges – in apparent image texture in the scales. Higher values in scale 3 suggest billboard like elongated Gaussians perpendicular to the scene surface that existing Generalizable 3DGS approaches produce. On the other hand our approach uses 2DGS rasterizer by construction forces this scale to be 0 eliminating this billboard like Gaussians. Not only the propose approach localize scene surfaces but the dominate structural edges are clearly visible in the estimated scales. For example, look at interesting case of the edge of the wall at the intersection of two corridors **??**. You can see that the Gaussians estimated by the proposed methods are very skewed in one direction than the other (one scale is very large and another very small). This is expected out of geometrically consistent Gaussians to have a very narrow extent perpendicular to geometric edges whereas an arbitrarily large extent along the edge. You also can see clearly that more planar areas have more homogenously elongated Gaussians. Whereas not as clear geometric edge or homogeneity in the size and shape of Gaussians can be found in the existing approaches.

We can also see the opacities of the reconstructed gaussians. Large regions of the image have very low opacities pointing to many near dead Gaussians estimated by the existing Generalizable 3DGS methods. These low opacity Gaussians are often found in the regions of shadows, light sources and relexion suggesting that instead of modelling these variations with spherical harmonics as one would expect, the veiw dependent color is been modelled by changing opacities. In contrast it can be seen that proposed method consist of less number of near dead Gaussians and low opacity regions corresponding to image edges and shadows.

5 Experiments

Some notes on geometric evaluation: When we ideally have infinite views, we can extract the surface mesh and evaluate mesh reconstruction accuracy using metrics like Chamfer distance, typically on datasets like DTU. However, when we don’t have dense multi-view coverage but instead have sufficient 2.5D RGB-D images (e.g., ScanNet), an alternative approach is to select two context views and evaluate the rendered depth. This serves as a proxy to assess interpolation/extrapolation quality on novel views—effectively measuring how well the model generalises depth consistency in a zero-shot setting.

Ideally on a dataset like DTU where 360 degree coverage of the scene is available, one can extract the ground truth surface mesh and evaluate reconstruction accuracy with a metric like Chamfer distance as done in []. As our method reconstruct scenes with two overlapping views only it is not a viable option. We have to limit the evaluation to the observed part of the scene – which is defined by union of view-frustum of input views. We thus try two alternatives: (i) mask the unobserved parts of the scene in the ground truth surface for evaluation of Chamfer distance or (ii) utilize depthmaps of the observed views within the union of context view-frustum and report average depth error (2.5D) on all the test views as a proxy for 3D geometric accuracy of estimated splats.

5.1 Datasets

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5.2 Evaluation Metrics

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5.3 Baselines

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5.4 Implementation Details

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5.5 Results & Analysis

Quantitative, Qualitative, ...

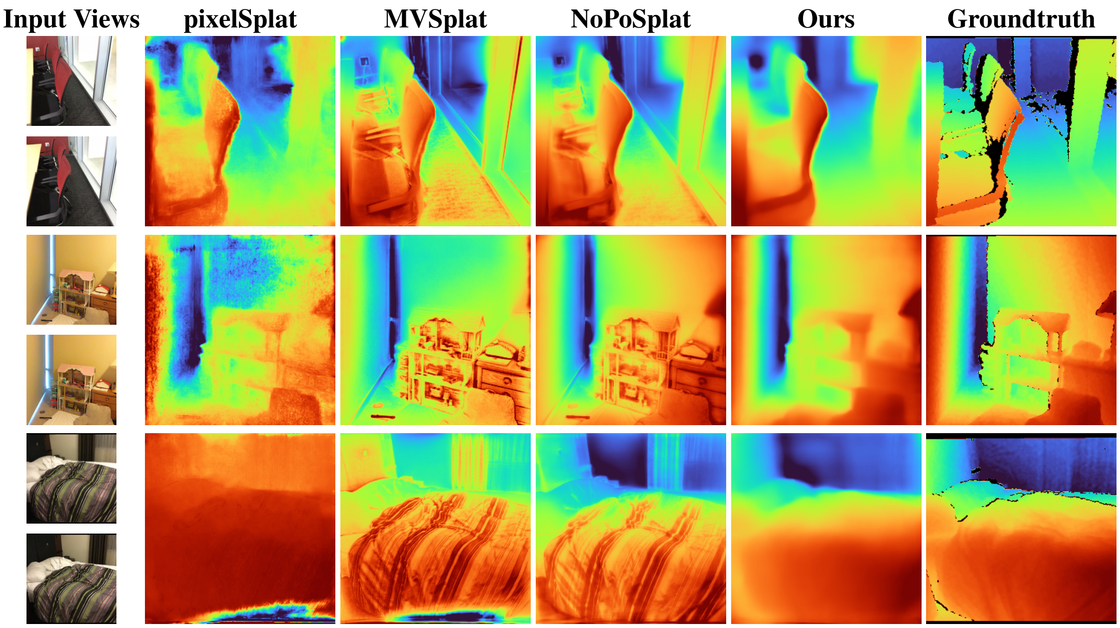

Figure 3: **Qualitative comparison of rendered novel-view depth on ScanNet-V1 [3].**

5.6 Ablations Studies

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6 Conclusion

Finally ...

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Declare funding (financial activities supporting the submitted work).

References

[1] David Charatan, Sizhe Lester Li, Andrea Tagliasacchi, and Vincent Sitzmann. pixelsplat: 3d gaussian splats from image pairs for scalable generalizable 3d reconstruction. In *CVPR*, 2024. 5

[2] Yuedong Chen, Haofei Xu, Chuanxia Zheng, Bohan Zhuang, Marc Pollefeys, Andreas Geiger, Tat-Jen Cham, and Jianfei Cai. Mvsplat: Efficient 3d gaussian splatting from sparse multi-view images. *arXiv preprint arXiv:2403.14627*, 2024. 5

[3] Angela Dai, Angel X Chang, Manolis Savva, Maciej Halber, Thomas Funkhouser, and Matthias Nießner. Scannet: Richly-annotated 3d reconstructions of indoor scenes. In *CVPR*, 2017. 8, 10, 11, 12

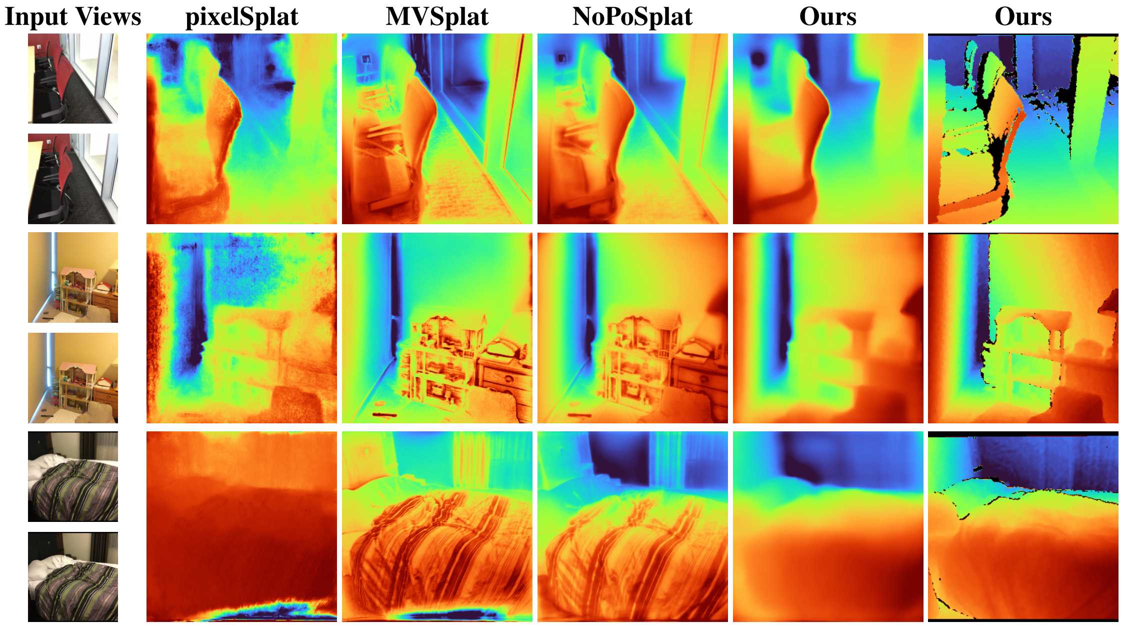

Figure 4: **Qualitative comparison of rendered novel-view depth on RE10k [7] (top two rows) and ACID [5] (bottom row).**

[4] Johan Edstedt, Qiyu Sun, Georg Bökman, Mårten Wadenbäck, and Michael Felsberg. Roma: Robust dense feature matching. In *CVPR*, 2024. 11

[5] Andrew Liu, Richard Tucker, Varun Jampani, Ameesh Makadia, Noah Snavely, and Angjoo Kanazawa. Infinite nature: Perpetual view generation of natural scenes from a single image. In *ICCV*, 2021. 9, 10, 11

[6] Shuzhe Wang, Vincent Leroy, Yohann Cabon, Boris Chidlovskii, and Jerome Revaud. Dust3r: Geometric 3d vision made easy. In *CVPR*, 2024. 5

[7] Tinghui Zhou, Richard Tucker, John Flynn, Graham Fyffe, and Noah Snavely. Stereo magnification: learning view synthesis using multiplane images. *ACM TOG*, 2018. 9, 10, 11

Table 1: **Novel-view synthesis performance on the RealEstate10k dataset [7].** We compare pose-required and pose-free methods against our proposed variants. "NoPoSplat" refers to results from publicly available checkpoints, while "NoPoSplat*" indicates results from our retrained baseline using identical hyperparameters for fair comparison. The overall best results among all methods are highlighted in **bold**, and the best-performing pose-free method (without optimizing camera poses for target views) is indicated by underline.

		Small			Medium			Large			Average		
	Method	PSNR↑	SSIM↑	LPIPS↓	PSNR↑	SSIM↑	LPIPS↓	PSNR↑	SSIM↑	LPIPS↓	PSNR↑	SSIM↑	LPIPS↓
Pose-Required	pixelNeRF	18.417	0.601	0.526	19.930	0.632	0.480	20.869	0.639	0.458	19.824	0.626	0.485
	AttnRend	19.151	0.663	0.368	22.532	0.763	0.269	25.897	0.845	0.186	22.664	0.762	0.269
	pixelSplat	20.263	0.717	0.266	23.711	0.809	0.181	27.151	0.879	0.122	23.848	0.806	0.185
	MVSplat	20.353	0.724	0.250	23.778	0.812	0.173	27.408	**0.884**	**0.116**	23.977	0.811	0.176
Pose-Free w/o Align.	NoPoSplat	21.086	0.721	0.237	23.134	0.776	0.185	25.086	0.818	0.141	23.189	0.775	0.185
	NoPoSplat*	21.097	0.723	0.237	23.191	0.779	0.187	25.107	0.817	0.144	23.244	0.778	0.187
	Ours (2DGS)	21.051	0.725	0.245	23.338	0.784	0.187	25.541	0.830	0.140	23.407	0.782	0.188
	Ours (2DGS+Grid)	21.341	0.736	0.236	23.453	0.788	0.185	25.529	0.829	0.141	23.528	0.787	0.185
	Ours (2DGS+Normal)	21.275	0.733	0.236	23.198	0.780	0.187	25.205	0.823	0.144	23.300	0.781	0.187
	Ours (2DGS+Grid+Normal)	21.349	0.739	0.234	23.463	0.789	0.184	25.628	0.832	0.141	23.564	0.789	0.184
Pose-Free w/ Align.	DUSt3R	14.101	0.432	0.468	15.419	0.451	0.432	16.427	0.453	0.402	15.382	0.447	0.432
	MASt3R	13.534	0.407	0.494	14.945	0.436	0.451	16.028	0.444	0.418	14.907	0.431	0.452
	Splatt3R	14.352	0.475	0.472	15.529	0.502	0.425	15.817	0.483	0.421	15.318	0.490	0.436
	CoPoNeRF	17.393	0.585	0.462	18.813	0.616	0.392	20.464	0.652	0.318	18.938	0.619	0.388
	NoPoSplat	22.514	0.784	**0.210**	24.899	0.839	**0.160**	27.411	0.883	0.119	25.033	0.838	**0.160**
	NoPoSplat*	**22.533**	**0.785**	0.211	**24.905**	**0.839**	0.161	**27.419**	**0.884**	0.120	**25.036**	**0.839**	0.161
	Ours (2DGS)	22.211	0.774	0.224	24.715	0.835	0.166	27.209	0.880	0.124	24.813	0.833	0.168
	Ours (2DGS+Grid)	22.376	0.780	0.217	24.733	0.836	0.166	27.228	0.880	0.125	24.869	0.835	0.167
	Ours (2DGS+Normal)	22.432	0.784	0.214	24.757	0.837	0.165	27.136	0.879	0.125	24.867	0.836	0.166
	Ours (2DGS+Grid+Normal)	22.423	0.784	0.215	24.787	0.838	0.165	27.282	0.882	0.124	24.941	0.838	0.165

Table 2: **Zero-shot out-of-distribution novel-view synthesis performance on ACID [5] and ScanNet-V1 [3], compared against state-of-the-art method.** All models are trained exclusively on the RE10k dataset. The best overall results are highlighted in **bold**, while the best-performing pose-free method (without optimizing camera poses for target views) is indicated by underline.

		ACID			ScanNet-V1		
	Method	PSNR↑	SSIM↑	LPIPS↓	PSNR↑	SSIM↑	LPIPS↓
w/o Eval. Align.	NoPoSplat	23.379	0.683	0.238	21.068	0.646	0.268
	Ours (2DGS)	23.914	0.704	0.233	21.167	0.648	0.267
	Ours (2DGS+Grid)	23.745	0.697	0.235	20.919	0.644	0.271
	Ours (2DGS+Normal)	23.848	0.702	0.235	21.178	0.648	0.271
	Ours (2DGS+Grid+Normal)	23.950	0.707	0.233	21.163	0.648	0.268
w/ Eval. Align.	NoPoSplat	**25.765**	**0.776**	**0.198**	**29.002**	**0.876**	**0.145**
	Ours (2DGS)	25.627	0.773	0.202	28.728	0.868	0.145
	Ours (2DGS+Grid)	25.629	0.772	0.202	28.575	0.864	0.148
	Ours (2DGS+Normal)	25.618	0.772	0.203	28.769	0.869	0.149
	Ours (2DGS+Grid+Normal)	25.734	**0.776**	0.201	28.984	0.873	**0.145**

A Appendix / supplemental material

Optionally include supplemental material (complete proofs, additional experiments and plots) in appendix. All such materials **SHOULD be included in the main submission.**

Table 3: **Pose estimation performance (AUC) across multiple error thresholds on RE10k [7] (in-domain), ScanNet-V1 [3], and ACID [5] (both cross-domain).** Our method achieves state-of-the-art results across all thresholds and datasets, despite not being trained on ScanNet or ACID. Remarkably, our approach significantly outperforms the baseline on the cross-domain ScanNet dataset using only PnP+RANSAC, without relying on additional photometric-based pose refinement, whereas the baseline depends on such optimization. The best overall results are highlighted in **bold**, and the best-performing PnP+RANSAC only method is indicated by <u>underline</u>. Note that RoMa [4], in contrast to our method, is directly trained on ScanNet.

		RE10k			ScanNet-V1			ACID		
	Method	5° ↑	10° ↑	20° ↑	5° ↑	10° ↑	20° ↑	5° ↑	10° ↑	20° ↑
	CoPoNeRF	0.161	0.362	0.575	-	-	-	0.078	0.216	0.398
	DUSt3R	0.301	0.495	0.657	-	-	-	0.166	0.304	0.437
	MASt3R	0.372	0.561	0.709	-	-	-	0.234	0.396	0.541
	RoMa	0.546	0.698	0.797	**0.168**	**0.361**	**0.575**	0.463	0.588	0.689
PnP+RANSAC only	NoPoSplat	0.572	0.728	0.833	0.078	0.198	0.394	0.337	0.497	0.646
	Ours (2DGS)	0.588	0.737	0.832	0.085	0.223	0.432	0.344	0.513	0.659
	Ours (2DGS+Grid)	0.621	0.760	0.849	0.123	0.279	0.471	0.382	0.540	0.674
	Ours (2DGS+Normal)	0.613	0.756	0.848	0.118	0.267	0.460	0.376	0.537	0.673
	<u>Ours (2DGS+Grid+Normal)</u>	<u>0.627</u>	<u>0.766</u>	<u>0.855</u>	<u>0.135</u>	<u>0.289</u>	<u>0.479</u>	<u>0.392</u>	<u>0.547</u>	<u>0.679</u>
w/ Photometric Optimisation	NoPoSplat	0.672	0.791	0.868	0.109	0.256	0.463	0.456	0.593	0.705
	Ours (2DGS)	0.672	0.788	0.859	0.129	0.298	0.515	0.460	0.599	0.713
	Ours (2DGS+Grid)	0.686	0.799	0.870	0.136	0.311	0.512	0.474	0.607	0.718
	Ours (2DGS+Normal)	0.679	0.798	0.871	0.141	0.323	0.520	0.475	0.610	0.721
	Ours (2DGS+Grid+Normal)	**0.689**	**0.804**	**0.876**	0.156	0.334	0.539	**0.488**	**0.619**	**0.726**

Table 4: **Pose estimation performance (AUC) across multiple error thresholds on ScanNet-1500 [3] (cross-domain).** Our method achieves state-of-the-art results across all thresholds despite not being trained on ScanNet. The best overall results are highlighted in **bold**, and the best-performing PnP+RANSAC only method is indicated by <u>underline</u>. Note that RoMa [4], in contrast to our method, is directly trained on ScanNet.

		ScanNet-1500		
	Method	5° ↑	10° ↑	20° ↑
	DUSt3R	0.221	0.437	0.636
	MASt3R	0.159	0.359	0.573
	RoMa	**0.271**	0.476	0.653
PnP+RANSAC only	NoPoSplat	0.232	0.443	0.634
	Ours (2DGS)	0.237	0.451	0.637
	Ours (2DGS+Grid)	0.238	0.467	0.657
	Ours (2DGS+Normal)	0.242	0.471	0.659
	<u>Ours (2DGS+Grid+Normal)</u>	<u>0.250</u>	<u>0.476</u>	<u>0.662</u>
w/ Photometric Optimisation	NoPoSplat	0.262	0.474	0.651
	Ours (2DGS)	0.241	0.462	0.650
	Ours (2DGS+Grid)	0.261	0.473	0.658
	Ours (2DGS+Normal)	0.255	0.480	0.657
	Ours (2DGS+Grid+Normal)	0.263	**0.479**	**0.666**

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Please read the checklist guidelines carefully for information on how to answer these questions. For each question in the checklist:

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Table 5: **Depth estimation performance for novel views on ScanNet-V1 [3].** Two input camera views are used to reconstruct the scene using Gaussians. We propose using the accuracy of rendered depth for novel views as a holistic measure of the model's interpolation capability for 3D reconstruction. Our method outperforms all other approaches across all evaluated metrics by clear margin. The best overall results are highlighted in **bold**, and the best-performing method without camera pose refinement is indicated by <u>underline</u>.

		Rendered Depth (Novel views)		
	Method	Abs Rel↓	$\delta_1 < 1.10 \uparrow$	$\delta_1 < 1.25 \uparrow$
	pixelSplat	0.299	0.552	0.818
	MVSplat	0.189	0.412	0.745
w/o *Pose* Refine.	NoPoSplat	0.131	0.554	0.851
	NoPoSplat*	0.135	0.552	0.847
	Ours (2DGS)	0.121	0.668	0.879
	Ours (2DGS+Grid)	0.109	0.679	0.890
	Ours (2DGS+Normal)	0.115	0.674	0.884
	Ours (2DGS+Grid+Normal)	<u>0.108</u>	<u>0.680</u>	<u>0.892</u>
w *Pose* Refine.	NoPoSplat	0.126	0.567	0.861
	NoPoSplat*	0.130	0.564	0.856
	Ours (2DGS)	0.114	0.692	0.884
	Ours (2DGS+Grid)	0.102	0.706	0.901
	Ours (2DGS+Normal)	0.106	0.704	0.898
	Ours (2DGS+Grid+Normal)	**0.100**	**0.707**	**0.904**

Table 6: **Depth estimation performance for input views on ScanNet-V1 [3].** The scene is reconstructed from two input camera views using Gaussians, and the reported "rendered depth" is computed directly from these views. The first input view serves as the global reference frame. The best overall results are highlighted in **bold**, and the best-performing method without camera pose refinement is indicated by <u>underline</u>.

		Input 1 Rendered Depth			Input 2 Rendered Depth		
	Method	Abs Rel↓	$\delta_1 < 1.10 \uparrow$	$\delta_1 < 1.25 \uparrow$	Abs Rel↓	$\delta_1 < 1.10 \uparrow$	$\delta_1 < 1.25 \uparrow$
w/o *Pose* Refine.	NoPoSplat	0.127	0.575	0.864	0.138	0.530	0.833
	NoPoSplat*	0.129	0.573	0.864	0.139	0.529	0.834
	Ours (2DGS)	0.110	0.701	0.892	0.127	0.628	0.858
	Ours (2DGS+Grid)	0.103	0.710	0.904	0.119	0.633	0.868
	Ours (2DGS+Normal)	0.105	0.709	0.898	0.123	0.632	0.863
	Ours (2DGS+Grid+Normal)	<u>0.101</u>	<u>0.713</u>	<u>0.906</u>	<u>0.118</u>	<u>0.637</u>	<u>0.870</u>
w *Pose* Refine.	NoPoSplat	0.127	0.575	0.864	0.122	0.572	0.865
	NoPoSplat*	0.129	0.573	0.864	0.123	0.569	0.865
	Ours (2DGS)	0.110	0.701	0.892	0.100	0.709	0.896
	Ours (2DGS+Grid)	0.103	0.710	0.904	0.097	0.717	0.903
	Ours (2DGS+Normal)	0.105	0.709	0.898	0.099	0.716	0.902
	Ours (2DGS+Grid+Normal)	**0.101**	**0.713**	**0.906**	**0.095**	**0.719**	**0.908**

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